Binary Lifting

"Concepts & Qns"

Video- 1

→ codestorywithmik

→ CswithMIK

→ codestorywithMIK

Motivation .

Refuse to be the person who only almost made it.
Drop the excuses and finish what you started.

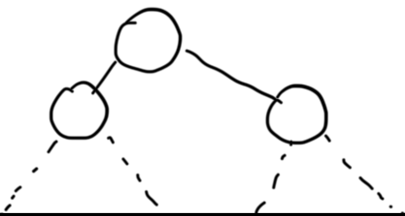
MIK...

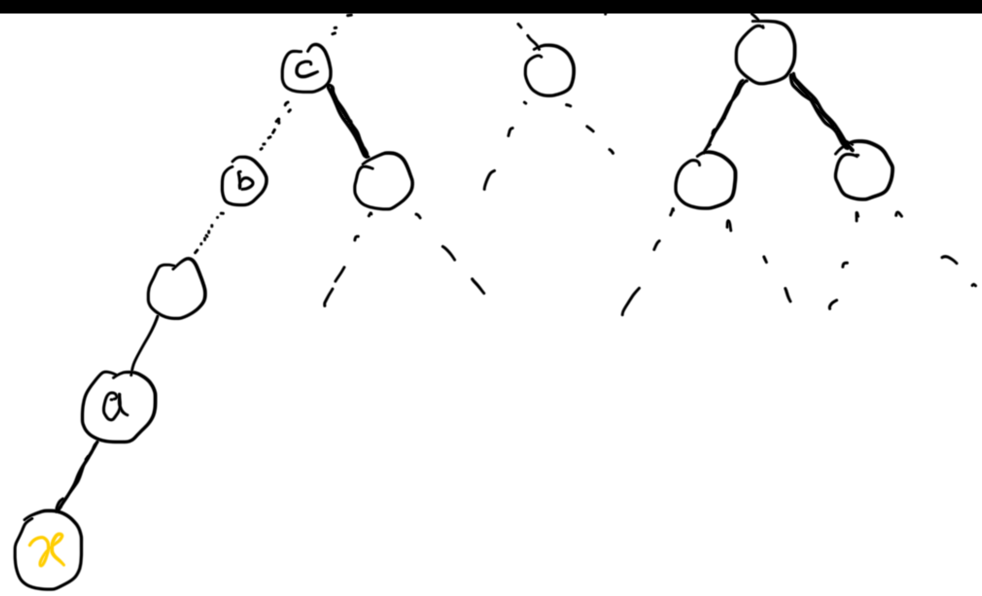
Introduction

What ? Why ? How ?

Binary Lifting....

Suppose I have a tree of ~2 lakh nodes.





- What is the ancestor of $\boxed{x}$? // a
- what is the 50th ancestor of x ? // b
- what is the 80th ancestor of x ? // c

K^{th} ancestor of node x

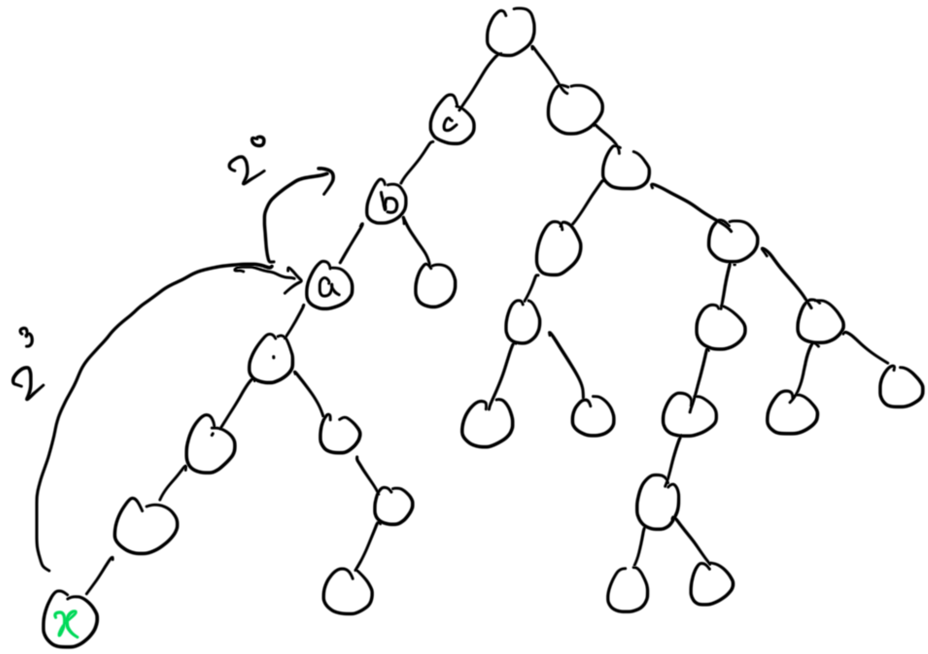
Time Complexity = $\underbrace{O(K)}_{\text{Slow}}$ Per query

Binary Lifting Comes to picture

- It helps to jump upward in the tree very efficiently.

→ Jump one step at a time ✗

→ climb in larger jumps ✓



9

$9 = 2^3 + 2^0$
$= 8 + 1$
$= 9$

2^0
2^1
2^2
$2^3 \to 8$
⋮

Jump in Power of 2.
i.e $2^0, 2^1, 2^2, 2^3 \ldots\ldots$

$$\boxed{x \to 2^i \to \text{which ancestor ???}}$$

Why Power of 2 only?

$x \rightsquigarrow$ 14 jumps $= \boxed{2^3} + \boxed{2^2} + \boxed{2^1}$

$= 3^3 + 3^1 + \boxed{3^0} + \boxed{3^0}$

Every no can be expressed as unique Power of 2.

$14 \rightarrow 1\ 1\ 1\ 0$

$$\begin{array}{cccc} 1 & 1 & 1 & 0 \\ 2^3 & 2^2 & 2^1 & 2^0 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 8 & +\ 4 & +\ 2 & +\ 0 \end{array} = 14$$

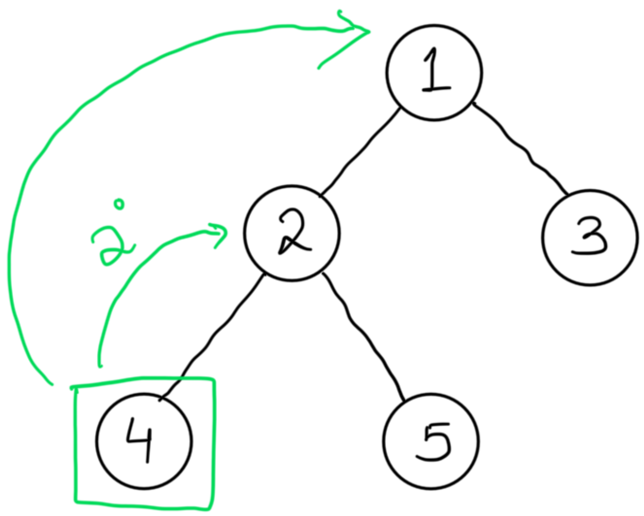
$13 \rightarrow 2^3 + 2^2 + 2^0$

$$\begin{array}{cccc} 1 & 1 & \cancel{0} & 1 \\ 2^3 & 2^2 & \cancel{2^1} & 2^0 \end{array}$$

Binary Lifting ...

How to store Power of 2 Jumps Information

$x \rightarrow 2^i \rightarrow$ which ancestor ???



$2^0 = 1$

$2^1 = 2$

$$4 \to 2^0 \to 2$$

$$4 \to 2^1 \to 1$$

node   Power   ancestor node

$$\text{parent}[4][0] = 2$$

$$\text{parent}[4][1] = 1$$

$$\boxed{\text{parent}[\text{node}][j] = 2^j\text{th ancestor node}}$$

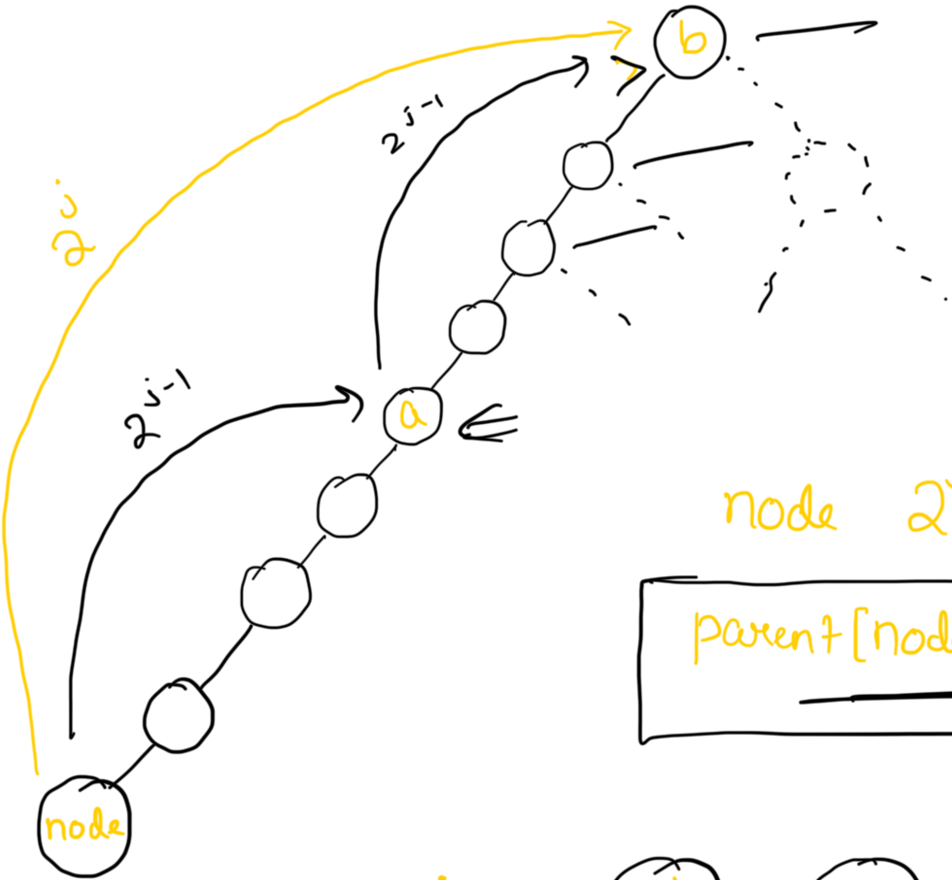
If i take 2^j jumps from node, which ancestor I will reach

$$\text{parent}[\text{node}][\overset{j=0}{0}] = 2^0\text{th ancestor} = 1 \text{ Jump up}$$

$$\text{parent}[\text{node}][\overset{j=1}{1}] = 2^1\text{th ancestor} = 2 \text{ Jumps up}$$

$\vdots$

The only thing to Keep in mind



node 2^j → ancestor ???

$$\boxed{\text{parent[node][j]} = ???}$$

$$2^j = 2^{j-1} + 2^{j-1}$$

a = [node][j-1]

b = parent[a][j-1]

Source

$$parent[node][j] = parent[a][j-1]$$

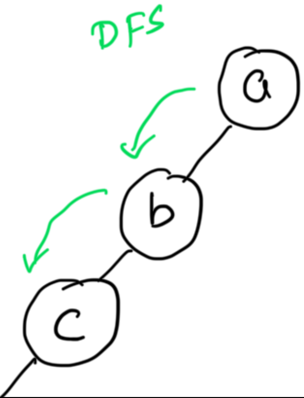
$$= parent[parent[node][j-1]][j-1]$$

$$parent[node][j] = parent\big[parent[node][j-1]\big][j-1]$$

Most Important Recurrence Relation (DP on Trees)

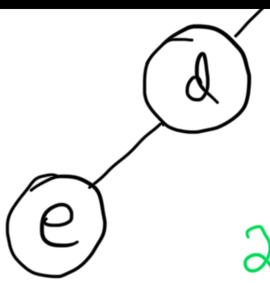
$n=5$

$\log_2 5 \simeq 2.$



$$parent[\underset{\text{rows}}{node}][\;j\;]$$

j: 0	j: 1	j: 2



$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

a	-1	-1	-1
b	a	-1	-1
c	b	a	-1
d	c	b	-1
e	d	c	a

$$\text{parent}[e][\overset{j=2}{2}] = \underline{\text{parent}[\;c\;][1]} = a$$

$$2^j = n \rightarrow \text{no of nodes}$$

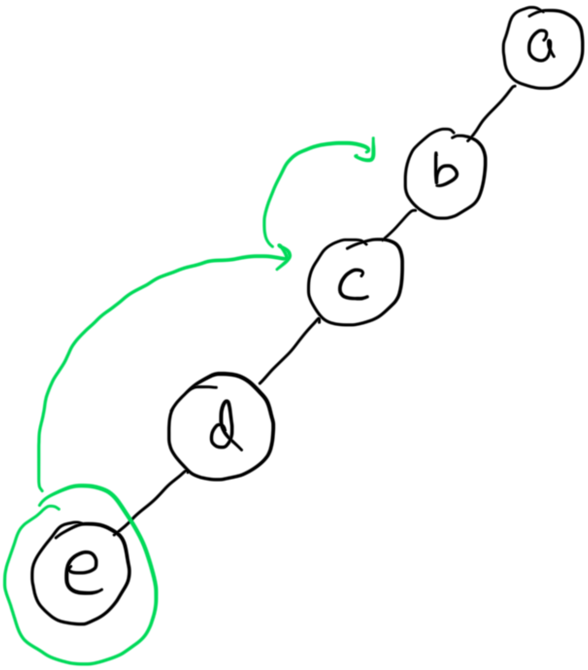
$$j = \log_2 n$$

$$\text{parent}[\;][\;] \begin{cases} \rightarrow \text{Rows} = \text{nodes} \\ \rightarrow \text{Columns} = \log_2(n)+1 \end{cases}$$

// Building the Ancestor Table.

```
for( j=1 ; j< log2(n)+1 ; j++) {
    for(node = 0; node <=n-1; node ++) {
        if( parent[node][j-1] != -1) {
```

```
parent[node][j] = parent[parent[node][j-1]][j-1];
            }
        }
    }
```



	0	1	2
a	-1	-1	-1
b	a	-1	-1
c	b	a	-1
d	c	b	-1
e	d	c	a

Find the 3rd ancestor of node = e

$$3 = 0\underset{2^1}{1}\underset{2^0}{1} = 2^1 + 2^0$$

c = parent[e][1]

b = parent[c][0]

Find K^{th} Ancestor

using Binary Lifting

Next $\Rightarrow$